

Predicting the Next Construction Boom

Forecasting Sector Level Labor Demand for Lithko Contracting

Final Paper

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Lithko Contracting

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1. Executive Summary:

Lithko Contracting operates in a construction services market where demand shifts across project types over time. For example, warehouses saw strong growth during COVID, while data centers are driving demand more recently. In most cases, these shifts have been recognized only after activity has already increased. While Lithko has strong internal data, there is an opportunity to combine it with external economic signals to identify these changes earlier and support more proactive planning.

In this project, we built a quarterly forecasting framework that integrates Lithko's internal labor-hours data with external macroeconomic indicators from sources such as the Federal Reserve Economic Data (FRED) and the U.S. Census Bureau. The idea was to compare a baseline model, which relies only on historical labor trends and a COVID adjustment, with augmented models that also include external indicators at different forecast horizons.

We applied this framework across five key sectors: Data Center, Warehouse, Apartment, Office, and Manufacturing and evaluated performance across six forecast horizons, from 1 quarter ahead to 6 quarters ahead. In total, we tested over 120 model variations. To keep the evaluation consistent and realistic, we used a time-based validation approach, where the most recent four quarters of 2025 were held out as a test period. We also followed a predefined model selection rule across all sectors and horizons to ensure the results were fair and not influenced by hindsight.

Key Findings:

Out of the five sectors we analyzed, four showed clear and actionable forecasting signals at horizons that are meaningful for planning:

- **Manufacturing** showed strong results at a 5-quarter horizon, with a 5.1% out-of-sample MAPE and an 83.8% improvement over the baseline. The most useful signal came from Private Nonresidential Fixed Investment (PNFI).
- **Data Center** also performed well at a 5-quarter horizon, with a 7.5% MAPE and an 80.1% lift over the baseline. In this case, a combination of indicators worked best, including data center capex, utility production, capacity utilization, and the PCE deflator.
- **Apartment** showed its best performance at a 4-quarter horizon, with a 9.7% MAPE and a 23.7% lift. The key indicator here was multifamily housing units under construction (undcon5musa).
- **Warehouse** produced results at a longer 6-quarter horizon, with a 21.2% MAPE and a 39.7% lift over the baseline. Similar to Data Centers, a combination of indicators performed best, including e-commerce sales, PNFI, warehouse construction activity, and RCPIX.
- **Office** was the only sector where we did not find a reliable signal at any planning-relevant horizon (3 to 6 quarters ahead). The shift toward remote and hybrid work after COVID appears to have broken the historical relationship between traditional office indicators and labor demand. Instead of forcing a result, we are reporting this as a clear limitation of the current data.

Strategic Implications For Lithko:

The framework provides Lithko with roughly one year to fifteen months of forward visibility on labor demand across four of its five major construction sectors. Given that Lithko's typical

approval and mobilization cycle is around two quarters, this level of lead time creates a real opportunity to plan ahead rather than react after demand has already increased. This can directly support better workforce planning, equipment allocation, and a more informed approach to managing the bid pipeline.

To make this useful in practice, we suggest a phased approach. First, automate the data pipeline so that external indicators update each quarter without manual work. Second, publish the sector-level forecasts for the four actionable segments so they are easily accessible for planning. Third, connect these forecasts to the bid review process, where certain thresholds can trigger early capacity discussions. Finally, revisit and revalidate the models each year as more data becomes available, and gradually expand into regional-level analysis once there is enough history to support it.

At the same time, it is important to be clear about the main limitation of this work. The dataset covers about 28 quarters from 2019 to 2025, which is relatively small for this type of modeling. Because of this, the results should be used as directional guidance rather than exact predictions. As more data is collected over time, the models can be strengthened and tested more rigorously.

2. Background and Industry Context:

2.1. About Lithko Contracting:

Lithko Contracting is a leading concrete subcontractor operating across the United States, with a strong focus on commercial and industrial construction. The company works across several project types, including data centers, warehouses, apartments, office buildings, and manufacturing facilities. This wide range of projects allows Lithko to closely observe how demand shifts across different parts of the nonresidential construction market.

A key part of Lithko's operations is aligning crews, equipment, and project management resources with incoming demand. When changes in the market are identified early, the company can plan ahead by positioning teams, preparing for bids, and allocating resources to the right regions and sectors. However, when these shifts are recognized only after activity has already picked up, Lithko is forced to react quickly, which can lead to higher costs and operational challenges.

2.2. Industry Context:

The U.S. nonresidential construction market has gone through noticeable shifts over the past decade, especially across different project types. During the COVID-19 period, the rapid growth of e-commerce led to a strong increase in warehouse construction between 2020 and 2022. More recently, starting around 2023, data center construction has grown significantly, mainly driven by the expansion of cloud computing and investments in artificial intelligence infrastructure.

At the same time, manufacturing construction has picked up due to policies like the CHIPS and Science Act and broader efforts to bring production back to the U.S. In contrast, office construction has slowed down, largely because remote and hybrid work have continued even after the pandemic.

Most of these shifts are influenced by larger economic trends that can be observed through publicly available data. The challenge, however, is figuring out which indicators actually give early signals for each construction segment, how far in advance they lead demand, and how reliable those signals are.

3. Problem Statement

Lithko Contracting has strong visibility into its current bid pipeline and ongoing projects, which provides a clear view of present activity across sectors. However, anticipating which construction segments will see growth in labor demand in future quarters remains a challenge. In many cases, shifts in demand become visible only after activity has already started to increase, which limits the ability to plan ahead.

While Lithko maintains detailed internal data on jobs, labor hours, customers, business units, and regions, this information has not yet been combined with external macroeconomic indicators in a structured forecasting approach. At the same time, publicly available data sources such as the Federal Reserve Economic Data (FRED) and the U.S. Census Bureau provide a wide range of indicators that are likely connected to construction activity.

This project focuses on understanding how these external indicators relate to Lithko's labor demand across different sectors. Specifically, the goal is to identify which indicators provide early signals, how far in advance those signals appear, and how reliable they are across different forecast horizons. An equally important part of the analysis is recognizing where the data does not support a consistent or actionable signal.

4. Project Description and Objectives

4.1. Project Objectives

This project focused on three main objectives:

- **Identify early signals:** Find macroeconomic indicators that consistently move ahead of changes in Lithko's labor demand for each construction sector.
- **Evaluate the improvement:** Understand whether combining Lithko's internal labor data with external economic indicators leads to better forecast accuracy compared to using only internal data.
- **Build a practical framework:** Develop a clear and repeatable forecasting approach that Lithko can run on a quarterly basis to support decisions.

4.2. Project Scope

This project focused on five construction sectors that represent a significant portion of Lithko's work: Data Center, Warehouse, Apartment (multifamily), Office, and Manufacturing. For each of these sectors, we evaluated forecasts at horizons ranging from one to six quarters ahead. Since Lithko's typical approval and mobilization cycle is around two quarters, we treated three quarters ahead as the minimum horizon that is useful for planning.

The analysis used data from 2019 Q1 through 2025 Q4, covering a total of 28 complete quarters. Internal data was sourced from Lithko's production systems and included job records, labor-hour transactions, customer details, business unit assignments, and regional information. External data was collected from sources such as the Federal

Reserve Economic Data (FRED) and the U.S. Census Bureau, with indicators selected based on their expected relevance to each construction sector.

5. Data Sources

5.1. Lithko Internal Data

Lithko's internal data for this project was sourced from several production database tables that capture different aspects of operations, including jobs, labor hours, customers, business units, and regions. The main tables used in this analysis were:

- Job-level details
- Daily labor-hour records
- Customer information
- Sector and business-unit details
- Regional classifications

One important step in the data preparation process was defining the actual start of a project. Instead of using the system-recorded job creation date, we used the date of the first recorded labor hour for each job. This ensures that labor hours are assigned to the correct quarter based on when work actually began, rather than when the job was created in the system.

After that, labor hours were aggregated to a quarterly level for each sector. We also checked the most recent quarter for completeness and excluded any partial quarter to avoid underestimating activity in the analysis.

5.2. External Macroeconomic Indicators

External indicators used in this analysis were sourced from two well-established public data sources:

- Federal Reserve Economic Data (FRED), maintained by the Federal Reserve Bank of St. Louis
- U.S. Census Bureau Construction Statistics

For each construction sector, indicators were selected based on their expected relationship to demand. In other words, each variable was chosen because it is likely to move ahead of changes in labor demand for that specific segment, based on industry understanding and economic reasoning. The full list of indicators used for each sector is summarized in the Results section and provided in more detail in Appendix A.

5.3. Frequency Conversion

Most of the external indicators from FRED are available at a monthly level, while Lithko's labor data is analyzed at a quarterly level. To make them comparable, all monthly indicators were converted to quarterly values by taking the average of the three months within each quarter.

This same approach was applied consistently across all indicators and sectors, without any special adjustments for individual variables. For indicators that are already available at a quarterly level, such as some Census housing series, the original values were used directly.

6. Methodology

6.1. Forecasting Framework

Our approach compares two types of models for every sector and every forecast horizon: A baseline model and an augmented version that includes external indicators.

6.1.1. Baseline Model

The baseline model uses only Lithko's historical labor data, along with a control for the COVID period:

Equation: $y_t = \beta_0 + \beta_1 \cdot y_{t-h} + \beta_2 \cdot \text{COVID}_t + \epsilon_t$

Here, y_t represents labor hours in the target quarter, and y_{t-h} represents labor hours from h quarters earlier. The COVID variable is a simple indicator that takes the value 1 during 2020 (Q1 through Q4) and 0 otherwise.

This baseline is not just a simple or naive forecast. It already accounts for the natural pattern in labor demand over time as well as the disruption caused during the COVID period.

6.1.2. Augmented Models

On top of the baseline, we tested three variations that include external macroeconomic indicators:

- **Levels (Parsimonious):** This version adds one external indicator to the baseline. The indicator is selected based on which one has the strongest relationship with labor hours at that specific lag.
 - **Equation:** $y_t = \beta_0 + \beta_1 \cdot y_{t-h} + \beta_2 \cdot X_{t-h} + \beta_3 \cdot \text{COVID}_t + \epsilon_t$
- **Change-in (Parsimonious):** This model is similar to the Levels version, but instead of using the indicator value directly, it uses the change in the indicator. This helps in cases where both labor hours and the indicator are trending upward over time.
- **Full Model:** This version includes all candidate indicators for a sector, along with seasonal controls. It acts as a reference model to compare against the simpler versions.

6.1.3. Quarter Encoding

For the Full model, we also include seasonal effects using quarter-based indicators. This is done using the `factor()` function in R, which creates three dummy variables for Q2, Q3, and Q4, with Q1 acting as the reference. This means the model captures seasonal differences across quarters without assuming a simple upward or downward trend.

6.1.4. Lag Structure and Information Leakage

For each forecast horizon, we only use information that would have been available at that point in time. For example, when predicting five quarters

ahead, we use only the values from five quarters earlier for both labor hours and the indicators. We do not include multiple lags together. This ensures that the model does not accidentally use future information and keeps the forecasting process realistic and fair.

6.2. Validation Approach

We evaluate the models using a time-based hold-out approach. The training data includes all quarters from 2019 Q1 through 2024 Q4, and the most recent four quarters of 2025 are kept aside as the test period. Both the baseline and augmented models are trained on the same data and then used to generate forecasts for these four unseen quarters.

Model performance is measured using Mean Absolute Percentage Error (MAPE) on the test period.

Because we use lagged variables, the actual number of observations available for training varies slightly by sector and forecast horizon, typically ranging between about 22 and 27 observations. The test set remains fixed at four quarters.

While a rolling or walk-forward validation approach is commonly used in time-series problems, it was not practical here due to the limited data. Using that approach would reduce the training data even further at each step. Instead, the hold-out method allows us to keep as much data as possible for model fitting while still testing performance on unseen data.

6.3. Selection Rule

To make sure results are consistent and not influenced by hindsight, we applied the same model selection rule across all sectors and horizons. The process has two filters followed by a ranking step:

Filter 1: The augmented model must perform better than the baseline in terms of RMSE on the test data. This ensures that adding external indicators actually improves the forecast.

Filter 2: The augmented model must have a MAPE of 30% or lower. This threshold reflects a level of accuracy that is considered useful for planning decisions.

Selection: Among the models that pass both filters, we choose the one with the lowest MAPE. If there is a tie, we select the model with the longer forecast horizon, since it provides more lead time for planning.

If no model meets these conditions for horizons between 3 and 6 quarters ahead, the sector is reported as having no actionable signal.

6.4. Performance Metrics

We use two main metrics to evaluate model performance on the test data:

- **Mean Absolute Percentage Error (MAPE):** This measures the average forecast error as a percentage of the actual value. Since it is unit-free, it allows for easy comparison across different sectors and is used as the main reporting metric.

- **Root Mean Squared Error (RMSE):** This measures the average size of the error in terms of labor hours. It is used to compare models against the baseline and to calculate improvement.

To measure how much the augmented model improves over the baseline, we calculate: $\text{Lift} = (\text{Baseline RMSE} - \text{Augmented RMSE}) / \text{Baseline RMSE}$.

A positive lift means the augmented model provides better out-of-sample accuracy than the baseline.

7. Results

7.1. Sector-by-Horizon Performance Grid

The table below shows the out-of-sample MAPE for the best-performing augmented model at each sector and forecast horizon. For each sector, we highlight the horizon that was selected based on our model selection rule. Horizons of one and two quarters are not included in the final selection since they fall below Lithko's minimum planning horizon of three quarters.

Sector	1Q	2Q	3Q	4Q	5Q	6Q	Selected
Data Center	-	-	7.6%	>30%	7.5%	10.3%	5Q
Warehouse	-	-	21.9%	28.1%	27.6%	21.2%	6Q
Apartment	-	-	11.0%	9.7%	13.2%	12.5%	4Q
Office	-	-	>30%	>30%	>30%	>30%	-
Manufacturing	-	-	11.7%	19.6%	5.1%	10.8%	5Q

Selection rule: We choose the horizon with the lowest MAPE, as long as it is at or below the 30% threshold. If multiple horizons meet this condition with similar accuracy, we select the longer horizon since it provides more lead time for planning.

For most sectors, at least one horizon meets this standard and provides a usable signal. However, for the Office sector, none of the horizons between 3Q and 6Q achieve MAPE below 30%, even after testing additional model variations such as post-2020 adjustments and log transformations. As a result, Office is reported as having no actionable signal based on the current data.

7.2. Cross-Sector Summary

The principal results across the five tested sectors are summarized in the table below:

Sector	Horizon	Spec	Indicators	MAPE	Lift
Data Center	5Q Ahead	Full	DC capex + utility production + capacity utilization + PCE deflator	7.5%	80.1%
Warehouse	6Q Ahead	Full	e-commerce sales + PNFI + warehouse construction + RCPIX	21.2%	39.7%

Apartment	4Q Ahead	Levels	undcon5musa (multifamily under construction)	9.7%	23.7%
Office	-	-	No actionable signal at 3Q - 6Q under any tested specification	-	-
Manufacturing	5Q Ahead	Levels	PNFI (Private Nonresidential Fixed Investment)	5.1%	83.8%

7.3. Data Center

Data Center is one of the strongest results in this analysis. We are able to get about five quarters of forward visibility with a 7.5% out-of-sample MAPE, and the augmented Full model improves accuracy by 80.1% compared to the baseline. The model uses a combination of indicators including data center construction spending, utility production, capacity utilization, and the PCE price deflator, all at a five-quarter lag.

From a business perspective, this makes sense. Data center projects are highly capital-intensive and are usually planned well in advance. Companies like Microsoft, Google, and Meta commit large amounts of capital months before actual construction begins. Because of this, signals like capex spending, energy infrastructure, and overall capacity utilization tend to move earlier and give a lead on labor demand. Together, these indicators provide a strong and consistent signal that leads Lithko's labor demand by around five quarters.

In the regression results, the data center capex variable at lag five is positive and statistically significant at the 5% level, which confirms that it adds meaningful predictive value beyond just the past labor trend.

7.4. Manufacturing

Manufacturing is the cleanest and most stable signal among all sectors. We see about five quarters of forward visibility with a very low 5.1% MAPE, and the model improves accuracy by 83.8% over the baseline. Interestingly, this result comes from a very simple model using just one external indicator: PNFI (Private Nonresidential Fixed Investment), at a five-quarter lag.

The business logic here is straightforward. Manufacturing investments typically involve large capital commitments that are made well before construction starts. Policies like the CHIPS Act and the push toward reshoring have made these investments more visible at the macro level. PNFI captures these trends effectively, which is why a single indicator is enough to produce strong results.

In the regression, the PNFI variable is highly significant, with a p-value well below 0.001. This is the strongest relationship we observed between a macroeconomic indicator and labor demand across all sectors.

7.5. Apartment

For Apartments, we see about four quarters of forward visibility with a 9.7% MAPE and a 23.7% improvement over the baseline. The selected indicator is the Census series for multifamily housing units currently under construction (undcon5musa).

The logic is fairly intuitive. If a building is already under construction today, there is a predictable need for labor, including concrete work, in the coming quarters. This makes the indicator a natural leading signal at the four-quarter horizon.

However, one important point to note is that in the regression itself, this indicator is not statistically significant when included alongside the lagged labor term and the COVID adjustment. Most of the predictive power comes from the past labor pattern rather than the indicator alone.

Even so, we still report this model because it produces the lowest out-of-sample error while meeting our selection criteria. So the takeaway for Lithko is that the indicator helps, but the main driver of the forecast is still the underlying labor trend.

7.6. Warehouse

Warehouse provides the longest horizon among the reportable sectors, with six quarters of forward visibility. The model achieves a 21.2% MAPE and improves accuracy by 39.7% over the baseline. The selected model includes multiple indicators such as e-commerce sales, PNFI, nonresidential construction spending, and the 10-year Treasury rate.

This sector is more volatile compared to others. Warehouse construction includes a wide range of project types, and large one-time projects can have a big impact on the data. Because of this, the forecasts are noisier, and the error is higher compared to Data Center or Manufacturing.

Even with that, the improvement over the baseline is meaningful, and the six-quarter horizon gives Lithko valuable lead time for planning and bidding decisions.

We also tested a stricter outlier removal rule, which initially showed very strong results. However, that version removed some valid data points from the test period, which artificially improved accuracy. Since that would not reflect a realistic scenario, we chose not to use it. The reported results are the more conservative and reliable version.

7.7. Office

Office is the only sector where we did not find a usable forecasting signal. None of the models at horizons between three and six quarters achieved a MAPE below 30%, even after testing additional variations such as adding a post-2020 shift or applying a log transformation.

The reason for this is fairly clear. The shift toward remote and hybrid work after COVID has changed the behavior of the office construction market. Indicators that used to track office demand no longer show a consistent relationship with labor activity.

Because of this, we report Office as a “no actionable signal” sector. Instead of forcing a model that performs poorly, it is more useful to be transparent and say that this

segment should be managed based on current pipeline visibility rather than macro forecasting, at least until the market stabilizes.

8. Limitations

We acknowledge several limitations in this analysis so the results are interpreted in the right context.

8.1. Sample Size

The biggest constraint in this project is the amount of data available. We are working with 28 quarters of data from 2019 through 2025. After applying lags, the actual number of observations used for training drops to around 22 to 27, depending on the sector and forecast horizon.

At longer horizons, where the models include more variables, this leads to a relatively small number of observations compared to the number of parameters being estimated. Because of this, metrics like R-squared from the training data are not very reliable indicators of real performance. Instead, we focus on how the model performs on unseen data, using MAPE as the main metric.

8.2. Out-of-Sample Test Power

The test period includes only four quarters, which is enough to evaluate performance directionally but not enough for strong statistical comparisons between models. Because of this, while we report the best-performing model based on our selection rule, small differences in MAPE between nearby horizons should not be interpreted as meaningful differences in accuracy.

8.3. Single Cycle Coverage

The data covers the period from 2019 to 2025, which includes the COVID disruption and the recovery period. However, it does not include multiple full economic cycles. Which means that the relationships we observe between indicators and labor demand may change over time. To account for this, the models should be revisited and updated as more data becomes available.

8.4. Variance Attribution

In models that include multiple variables, the overall performance comes from how all variables work together. While we can test whether an individual indicator is statistically significant, it is not meaningful to assign a fixed portion of the model's explanatory power to any one variable. Instead, indicator significance should be interpreted as its contribution after accounting for the other variables in the model.

8.5. Mitigations Applied

To work within these constraints, we made a few deliberate choices:

- We preferred simpler models whenever they performed well, to avoid overfitting with too many variables.
- We focused on out-of-sample MAPE rather than in-sample metrics.

- We applied a strict MAPE threshold of 30% to ensure only planning-useful models were considered.
- We reported sectors like Office as having no actionable signal instead of forcing a weak result.

9. Recommendations

9.1. Sector Level Recommendations

9.1.1. Data Center

For the Data Center segment, Lithko should track data center construction spending and related capital investment indicators on a quarterly basis. The 5-quarter-ahead forecast provides strong visibility, so it can be used to plan ahead for hiring, training, and equipment needs.

A practical approach would be to trigger a capacity planning review whenever the forecast increases meaningfully compared to the previous quarter, for example by around 10%. This gives enough time to prepare before demand fully materializes.

9.1.2. Warehouse

For Warehouse, the focus should be on monitoring trends in e-commerce activity and broader investment indicators like PNFI. The 6-quarter forecast provides a longer-term view, but the error is higher compared to other sectors. Because of this, the forecast is better suited for strategic planning and bid positioning rather than short-term staffing decisions. It is also important to communicate the wider uncertainty range clearly when sharing these forecasts with operational teams.

9.1.3. Apartment

For Apartments, the key indicator is the Census series for multifamily housing under construction. Lithko can track this regularly and use the 4-quarter-ahead forecast to support planning decisions.

At the same time, it's important to recognize that most of the predictive strength comes from the underlying labor trend, with the external indicator providing a smaller but consistent improvement. So this forecast should be used as a supporting signal rather than the sole driver of decisions.

9.1.4. Office

For Office, the recommendation is to treat it as a reactive segment for now. Since the models did not produce a reliable signal, planning should rely more on current pipeline visibility, customer announcements, and market updates. At the same time, Lithko should continue to monitor this segment. As the post-pandemic market stabilizes, there may be an opportunity to revisit and rebuild forecasting models in the future.

9.1.5. Manufacturing

Manufacturing is the strongest and most reliable segment in this analysis. Lithko should actively track PNFI as a leading indicator and use the 5-quarter-ahead forecast as a key input for workforce planning. Given the low error and strong improvement over the baseline, this forecast can be used with a higher level of confidence compared to other sectors.

9.2. Implementation Roadmap

9.2.1. Quarter 1: Data Refresh Pipeline

The first step is to automate the data process. Right now, external data from FRED and the Census Bureau is downloaded manually. This can be converted into an automated pipeline that updates the data every quarter. The R Markdown file used in this project can also be scheduled to run automatically at the end of each quarter, so the forecasts are refreshed without manual effort.

9.2.2. Quarter 2: Forecast Dashboard

Once the data pipeline is in place, the next step is to make the forecasts accessible. The four actionable sectors Data Center, Warehouse, Apartment, and Manufacturing can be published in a simple dashboard or planning portal. Along with the forecast values, it is important to include the historical error range so that users understand how much uncertainty to expect when using these forecasts.

9.2.3. Quarter 3: Planning Integration

After the Forecast dashboard, the forecasts can be connected directly to planning workflows. For example, Lithko can define clear triggers such as a meaningful increase in a sector forecast, which would prompt a capacity planning discussion. Assigning clear ownership for reviewing and acting on these signals will help ensure that the forecasts are actually used in decision-making.

9.2.4. Year 1+: Annual Revalidation

As more data becomes available, the models can be updated and improved over time. With a longer history, it will also be possible to break the analysis down further by region, which can make the forecasts even more useful. In addition, once enough data is available, more advanced validation approaches such as rolling or walk-forward testing can be introduced to strengthen confidence in the results.

10. Out of Scope and Future Work

There are several areas that were outside the scope of this project but could add significant value if explored further in the future.

- **Extended Data Window:** The biggest improvement would come from having more historical data. With a longer time range, the models can be tested more reliably, and

it becomes possible to evaluate performance across multiple business cycles. It would also allow the use of more advanced validation approaches.

- **Regional Analysis:** This analysis was done at a national level, but construction activity can vary significantly by region. For example, growth patterns in the Sun Belt are often very different from the Northeast. Extending this work to a regional level could provide more precise and actionable signals.
- **Project Subtype Analysis:** Within each sector, different types of projects may behave differently. For example, data centers can include hyperscale or colocation projects, and warehouses can vary by use. Breaking the analysis down into these subcategories could provide more detailed insights.
- **Forward-Looking Indicators:** There are also other external indicators that specifically track future construction activity. For example, the Architecture Billings Index (ABI) and similar industry surveys are designed to signal upcoming demand. Including these could strengthen the forecasting framework.
- **Cost-Side Factors:** This analysis focuses on demand, but labor hours are also influenced by cost factors such as wages, material prices, and equipment availability. Including these variables could help separate true demand changes from cost-driven variations in labor usage.

11. Conclusion

This project shows that external economic indicators can provide useful forward visibility into Lithko's labor demand across several key construction sectors. For four out of the five sectors we analyzed, the models were able to identify signals far enough in advance to support decisions around workforce planning, equipment allocation, and bid strategy. The framework was also designed to be practical and repeatable, so it can be run each quarter using updated data, with results that follow a consistent and transparent selection approach.

Among the sectors, Manufacturing produced the strongest and most stable signal, with PNFI providing about five quarters of forward visibility at a low error level. Data Center also showed a strong signal using a combination of indicators, again at around five quarters ahead. Apartment provided a clear four-quarter signal based on publicly available housing data, while Warehouse offered a longer six-quarter view, although with higher variability. Office was the only sector where we did not find a reliable signal, largely due to the structural changes in the market after COVID.

At the same time, it is important to recognize the limitations of this work. The analysis is based on a relatively short data window, which limits the statistical strength of the results. Because of this, the framework should be used as a directional planning tool rather than a precise forecasting system. As more data becomes available, the models can be refined and tested more rigorously over time.

Finally, we would like to thank the Lithko team for their support and collaboration throughout this project, especially for their time, feedback, and engagement during the final presentation. We are also grateful to Professor Gorman for his guidance and for pushing us to strengthen the methodology. Their input played a key role in improving the quality of this work.

Appendix A: Indicator-to-Sector Mapping with Business Logic

The following table documents the external macroeconomic indicators tested for each sector, the data source, and the business rationale for selecting each indicator.

Sector	Indicator	Source	Business Logic
Data Center	DC construction spending	Census	Hyperscale data center builds are capex-heavy; capital commits many quarters before crews arrive on site.
	Utility production	FRED	Data centers require power infrastructure planned alongside construction; utility output reflects related buildouts.
	Total capacity utilization (TCU)	FRED	When existing industrial capacity is high, capex investment in additional capacity follows.
	PCE deflator	FRED	Inflation expectations affect the timing of capex commitments; relevant for capex-heavy DC builds.
Warehouse	E-commerce retail sales	Census	E-commerce growth drives demand for distribution real estate, which leads warehouse construction.
	PNFI	FRED	Broad commercial real-estate investment indicator capturing private nonresidential construction activity.
	Warehouse construction spending	Census	Direct measure of sector-level construction activity at the macro level.
	10-Year Treasury (DGS10)	FRED	Construction financing cost; affects project go/no-go decisions and the timing of capital deployment.
Apartment	undcon5musa	Census	Multifamily units physically under construction today need concrete crews approximately four quarters later.
	Multifamily permits (5+ units)	Census	Permits issued today translate to construction in 6–12 months; an upstream leading indicator.
	10-Year Treasury (DGS10)	FRED	Proxy for mortgage and construction financing rates; affects multifamily project economics.
Office	U.S. Private Business Services spending	Census	Reflects business activity that drives office demand; weak post-COVID due to hybrid work shift.
	Total nonresidential construction	Census	Broad indicator capturing aggregate nonresidential building activity.
	10-Year Treasury (DGS10)	FRED	Office construction financing cost.
	PNFI	FRED	Broad investment indicator; tested as a candidate office leading indicator without success.

Manufacturing	PNFI	FRED	Industrial capex commits 6–12 months before groundbreaking; CHIPS Act and re-shoring create large pipeline signals.
	Industrial production index	FRED	Existing plant utilization signals need for capacity additions.
	Total capacity utilization (TCU)	FRED	When industrial capacity is high, capex investment in additional capacity follows.
	10-Year Treasury (DGS10)	FRED	Manufacturing financing cost; affects capex deployment timing.

Appendix B: Selected-Spec Regression Coefficients

This appendix presents the regression outputs for the four sector–horizon specifications highlighted in the main results. The tables include coefficient estimates, standard errors, and significance levels, based on the final models from the unified U-section R Markdown analysis.

B.1 Data Center 5Q Ahead, Full Specification:

Model: labor_hours ~ labor_hours_lag5 + dc_spending_lag5 + utility_production_lag5 + tcu_lag5 + pce_deflator_lag5 + qtr + COVID

Training observations: 20 | Test observations: 4

In-sample R-squared: 0.898 | Out-of-sample MAPE: 7.5%

Term	Estimate	Std. Error	p-value	Sig.
(Intercept)	-2,042,631	962,471	0.060	.
labor_hours_lag5	-1.67	1.75	0.361	
dc_spending_lag5	0.0008	0.0003	0.012	*
pce_deflator_lag5	-4,085	4,049	0.337	
tcu_lag5	22,580	14,000	0.138	
utility_production_lag5	2,152	1,579	0.203	
qtrQ2	18,346	28,624	0.536	
qtrQ3	43,851	27,568	0.143	
qtrQ4	9,571	26,702	0.728	
COVID	4,703	41,484	0.912	

B.2 Warehouse 6Q Ahead, Full Specification:

Model: labor_hours ~ labor_hours_lag6 + ecom_sales_lag6 + pnfi_lag6 + tlhrescons_lag6 + indpro_lag6 + dgs10_lag6 + qtr + COVID

Training observations: 22 | Test observations: 4

In-sample R-squared: 0.927 | Out-of-sample MAPE: 21.2%

Term	Estimate	Std. Error	p-value	Sig.
(Intercept)	-295,800	1,027,689	0.779	
labor_hours_lag6	0.640	0.370	0.112	
ecom_sales_lag6	-3.27	1.83	0.102	
pnfi_lag6	1,730	768	0.046	*
tlInrescons_lag6	-0.884	0.955	0.374	
indpro_lag6	-26,485	20,910	0.232	
dgs10_lag6	-264,026	48,064	<0.001	***
qtrQ2	22,480	56,818	0.700	
qtrQ3	77,010	51,406	0.162	
qtrQ4	70,738	51,862	0.200	
COVID	-18,245	58,046	0.759	

B.3 Apartment 4Q Ahead, Levels Specification:

Model: labor_hours ~ labor_hours_lag4 + undcon5musa_lag4 + COVID

Training observations: 25 | Test observations: 4

In-sample R-squared: 0.851 | Out-of-sample MAPE: 9.7%

Term	Estimate	Std. Error	p-value	Sig.
(Intercept)	973,695	377,126	0.017	*
labor_hours_lag4	1.139	0.315	0.002	**
undcon5musa_lag4	-1,265	1,137	0.278	
COVID	-518,511	131,935	<0.001	***

B.4 Manufacturing 5Q Ahead, Levels Specification:

Model: labor_hours ~ labor_hours_lag5 + pnfi_lag5 + COVID

Training observations: 24 | Test observations: 4

In-sample R-squared: 0.805 | Out-of-sample MAPE: 5.1%

Term	Estimate	Std. Error	p-value	Sig.
(Intercept)	-419,167	89,954	<0.001	***
labor_hours_lag5	-0.180	0.341	0.603	
pnfi_lag5	177	34.0	<0.001	***
COVID	-52,925	24,037	0.040	*

Significance codes: *** $p < 0.001$ ** $p < 0.01$ * $p < 0.05$. $p < 0.10$

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